CO-4 Graph Theory

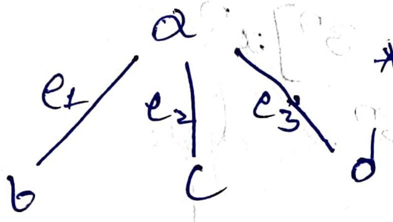
03/11/2

Graph:- A graph 'G' consist of a pair of (V, E) where 'V' is a non-empty finite set whose elements are called vertices and 'E' is another set whose elements are called edges. Such that each edge is associated with pair of vertices. Order of the graph = no of vertices.

* Size of the graph = |E|
* Order of the graph = |V|

Example:- 'V' = {a, b, c, d}

E = {(a,b)(a,c)(a,d)}

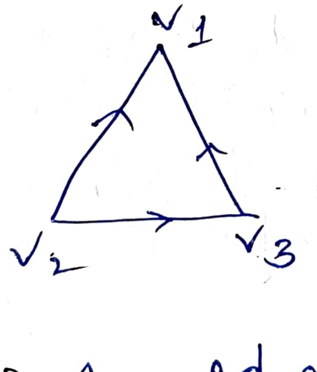


* It is a (4,3) graph.

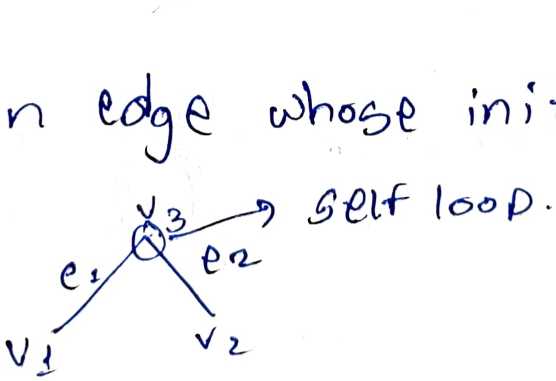
Finite graph:- Vertices and edges are countable

Undirected graph:- This is a two way direction graph.

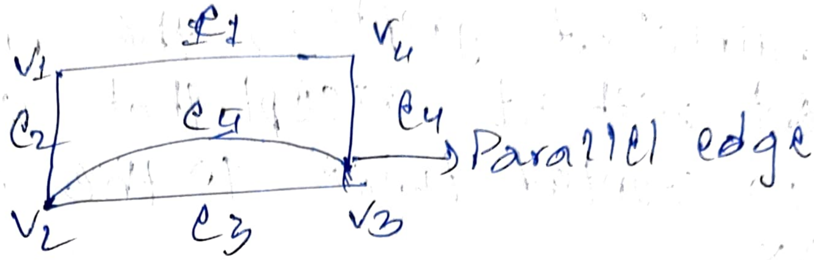
Directed graph:- This is one way direction graph i.e; each edge is given direction.



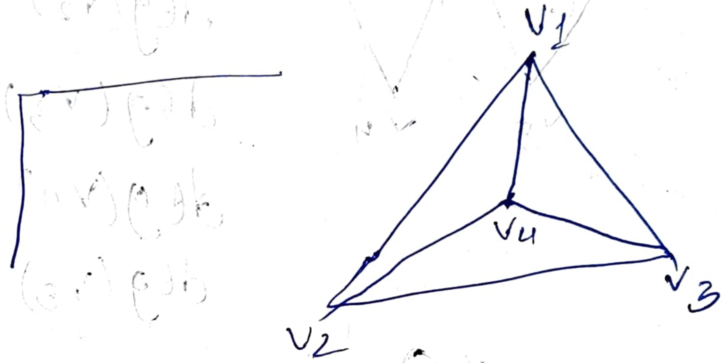
Self loop:- An edge whose initial & terminal vertex is same.



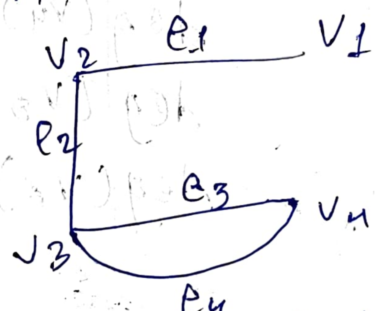
Parallel edge:- End edges of two vertices are same



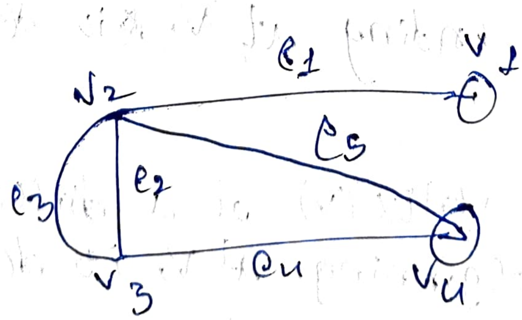
Simple Graph:- A graph with no loop and no multiple edges.



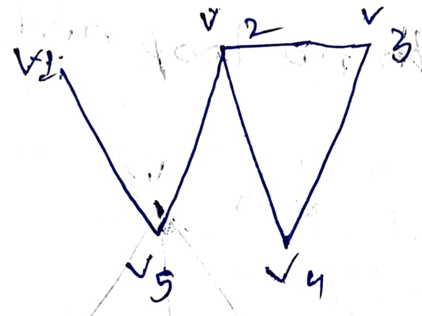
Multi Graph:- If more than one line joining with two vertices are alone in a graph is called multi graph.



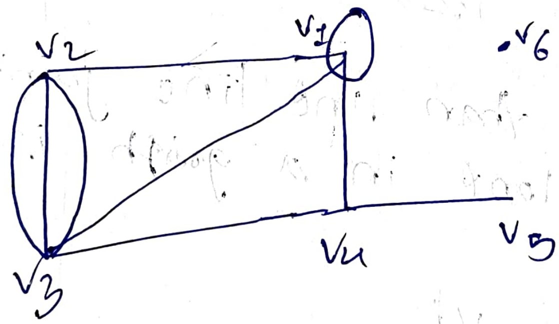
Psuedo Graph:- A graph in which loops and multi edges are alone is said to be psuedo graph.



Degree of a Vertex:- The degree of a vertex of an undirected graph is the number of edges incident with it except that a loop of a vertex counted twice to the degree of a vertex.



$deg(v_1) = 1$
$deg(v_2) = 3$
$deg(v_3) = 2$
$deg(v_4) = 2$
$deg(v_5) = 2$



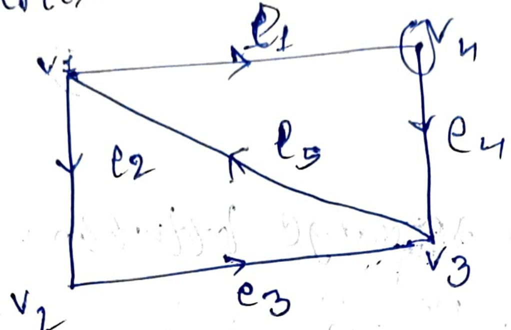
$deg(v_1) = 5$
$deg(v_2) = 4$
$deg(v_3) = 5$
$deg(v_4) = 3$
$deg(v_5) = 1$
$deg(v_6) = 0$

In-degree & Out-degree
$deg^-(v)$ $deg^+(v)$

* The In-degree of vertex (v) of a directed graph G is no of edges ending at v, is denoted by $deg^-(v)$.

* The Out-degree of vertex (v) of a directed graph G is no of edges beginning at v, is denoted by $deg^+(v)$.

* If the loop exists at any vertex then add 1 to the In & out degrees of that respective vertex.



$deg^-(v_1) = 1$

$deg^-(v_2) = 1$

$deg^-(v_3) = 2$

$deg^-(v_4) = 1+1$

$deg^+(v_1) = 2$

$deg^+(v_2) = 1$

$deg^+(v_3) = 1$

$deg^+(v_4) = 1+1$

Hand-Shaking Theorem (Fundamental Theorem)

* Sum of the degrees of vertices in an undirected graph is even. i.e; $\boxed{\sum_{i=1}^{all} deg(v_i) = 2|E|}$

un

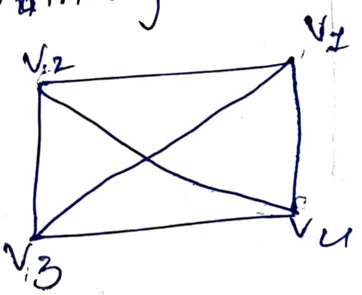
* For directed graph

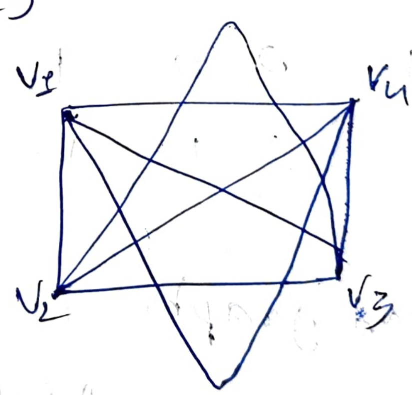
$$\boxed{\sum deg^-(v_i) = \sum deg^+(v_i) = |E|}$$

Complete Graph:-

A simple graph is said to be complete if every vertex in 'G' is connected with remaining all vertices

Ex:-

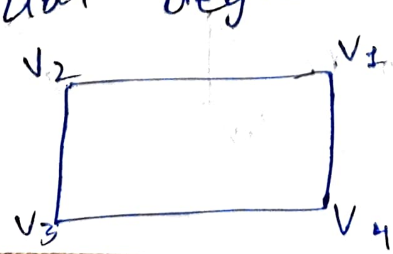




Regular Graph

A graph in which all the vertices are of equal degree.

Ex:-



$d(v_1) = 2$

$d(v_2) = 2$

$d(v_3) = 2$

$d(v_4) = 2$

Representation of Graphs by matrix.

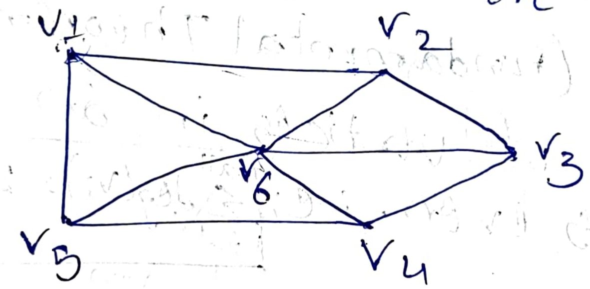
→ Adjacency matrix :-

* For undirected graph

$A = [a_{ij}]_{n \times n}$

where $a_{ij} = \begin{cases} 1 & \text{if there is an edge between } v_i \,\&\, v_j \\ 0 & \text{if there is no} \\ 2 & \text{self - loop} \end{cases}$

Ex:-

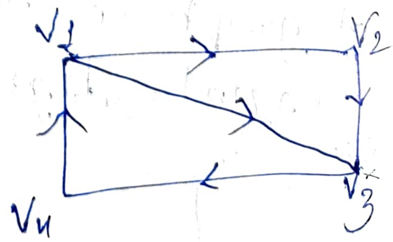


The Adjacency matrix of the graph is

$$\begin{array}{c|cccccc} & v_1 & v_2 & v_3 & v_4 & v_5 & v_6 \\ \hline v_1 & 0 & 1 & 0 & 0 & 1 & 1 \\ v_2 & 1 & 0 & 1 & 0 & 0 & 1 \\ v_3 & 0 & 1 & 0 & 1 & 0 & 1 \\ v_4 & 0 & 0 & 1 & 0 & 1 & 1 \\ v_5 & 1 & 0 & 0 & 1 & 0 & 1 \\ v_6 & 1 & 1 & 1 & 1 & 1 & 0 \end{array}$$

* For directed graph

The matrix of directed graph with 'n' vertices.

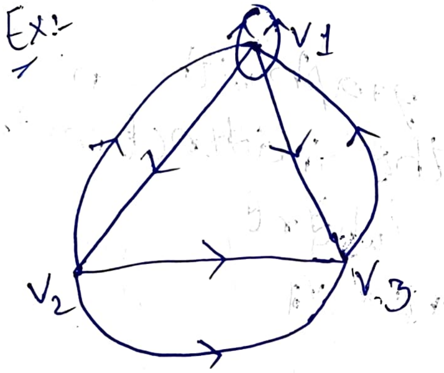


$A = [a_{ij}]_{n \times n}$

$a_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \text{ is in } D \\ 0 & \text{otherwise} \end{cases}$

$$\begin{array}{c|cccc} & v_1 & v_2 & v_3 & v_4 \\ \hline v_1 & 0 & 1 & 1 & 0 \\ v_2 & 0 & 0 & 1 & 0 \\ v_3 & 0 & 0 & 0 & 1 \\ v_4 & 1 & 0 & 0 & 0 \end{array}$$

Ex:-



$\Rightarrow$

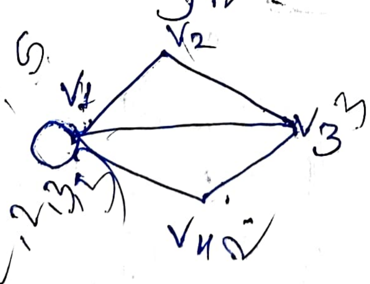
$$\begin{array}{c|ccc} & v_1 & v_2 & v_3 \\ \hline v_1 & 1 & 1 & 1 \\ v_2 & 1 & 0 & 2 \\ v_3 & 1 & 0 & 0 \end{array}$$

Degree sequence of graph 'G'

* $v_1, v_2, v_3, \ldots v_n$ are n-vertices of 'G' then the degree sequence $(d_1, d_2, \ldots d_n)$ where $d_i = \deg(v_i)$ is called the degree sequence of 'G'.

* In general we order the vertices so that the degree sequence is monotonically increasing.

Ex:-



(2,2,3,5)

$\deg(v_1) = 3+2 = 5$

$\deg(v_2) = 2$

$\deg(v_3)$ 3

$\deg(v_4) = 2$

The degree sequence of Graph is (2,2,3,5).

Note:- To construct graph the no of odd vertices must be even.

$(2, 2, \underline{3}, \underline{5})$

even no of odd vertices

$(\underline{1}, \underline{1}, 2, \underline{3}, 4, \underline{5})$.

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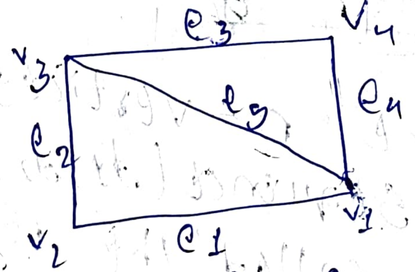
Incidence matrix

Let $G = (V, E)$ be an undirected graph with 'n' vertices and 'm' edges then the incidence matrix is given by matrix $I = [b_{ij}]_{n \times m}$ where

$$b_{ij} = \begin{cases} 1, & e_j \text{ is incident on the vertex } v_i \\ 0, & \text{otherwise.} \end{cases}$$

Ex:-

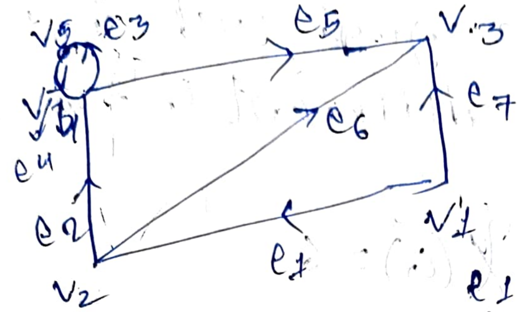


$$I_G = \begin{array}{c|ccccc} & e_1 & e_2 & e_3 & e_4 & e_5 \\ \hline v_1 & 1 & 0 & 0 & 1 & 1 \\ v_2 & 1 & 1 & 0 & 0 & 0 \\ v_3 & 0 & 1 & 1 & 0 & 1 \\ v_4 & 0 & 0 & 1 & 1 & 0 \end{array}$$

* For directed graph

$I_G = [b_{ij}]_{n \times m}$

$$b_{ij} = \begin{cases} 1, & e_j \text{ is incident out of the vertex } v_i \ (v_i \text{ is initial}) \\ -1, & \text{if } e_j \text{ is incident into the vertex } v_i \ (v_i \text{ is terminal}) \\ 0, & e_j \text{ is not incident on the vertex } v_i. \\ 1, & \text{if there is self-loop} \end{cases}$$

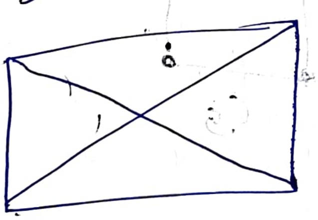


$$I_G = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array} \begin{array}{c} \begin{array}{ccccccc} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ -1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & -1 \\ 0 & -1 & & -1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Sub-Graph

→ A graph is obtained by deleting some edges or vertices.

Ex:-

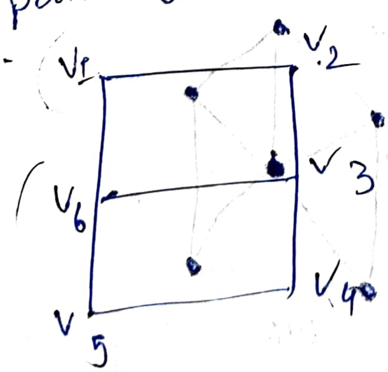


Sub graphs ⇒

Adjacent Vertices

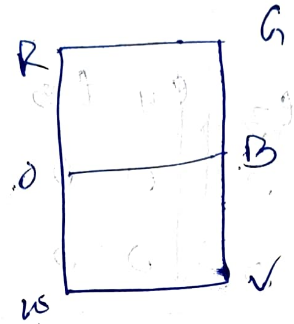
→ A pair of vertices connected by an edge.

Ex:-



(v_1, v_2)

→ If we assign colour to a graph such that the adjacent colours have different colours.



$\chi(G) = 6$

Chromatic Number:-

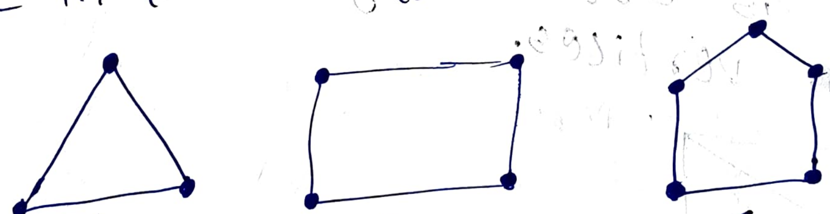
Let G be a graph the minimum number of colours assigned on adjacent vertices.

* Denoted by $\chi(G)$.

Cyclic graph (C_n).

→ The initial and terminal vertex are same

Ex:-

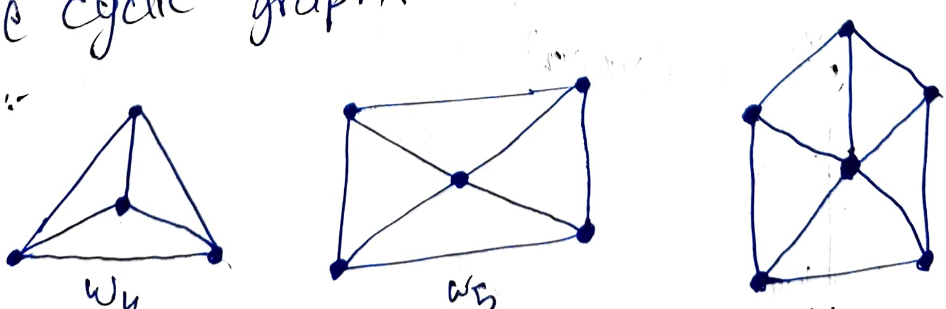


C_3 C_4 C_5

Wheel Graph (W_n)

→ The wheel graph with 'n' vertices is denoted by (W_n) obtained from cyclic graph by adding one vertex which is adjacent to all vertices in the cyclic graph.

Ex:-



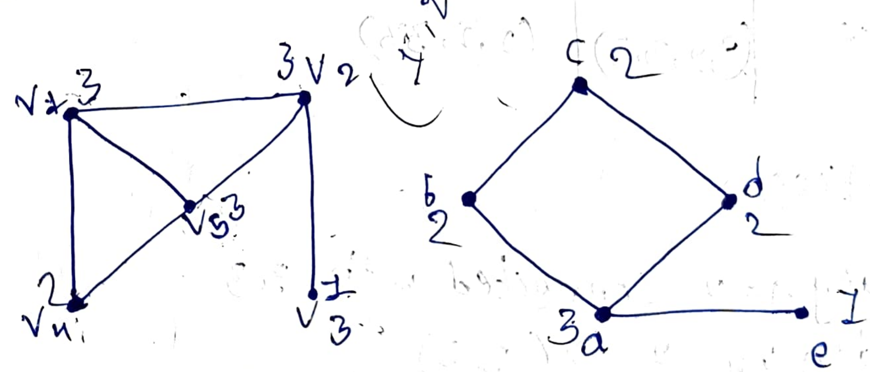
W_4 W_5 W_6

Isomorphism of Graphs

→ Two graphs G_1 & G_2 are isomorphic if and only if

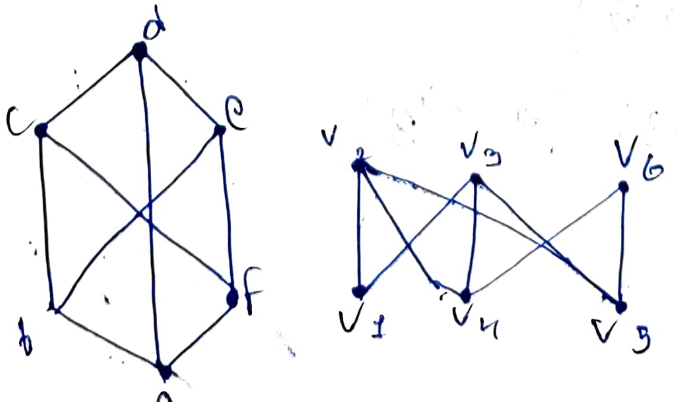
1. The no of vertices are same.
2. No of edges are same.
3. Degree sequence is same.
4. one-one correspondance for vertices & edges.
5. Adjacency matrix satisfies.

Ex:-



	G_1	G_2
No of vertices	5	5
No of edges	6	5

∴ No of edges are not equal so they are not isomorphic.

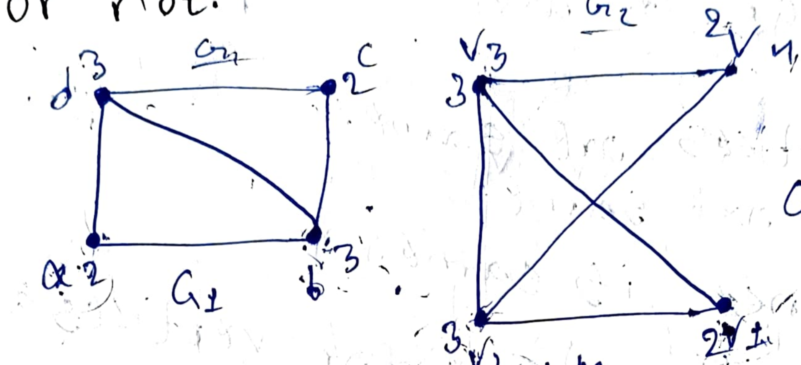


	G_1	G_2
No of vertices	6	6
No of edges	9	8

∴ Not isomorphic

Verify the given two graphs are isomorphic or not.

2, 3, 3



	G1	G2
No of vertices	4	4
No of edges	5	5
Degree sequence	(2, 2, 3, 3)	(2, 2, 3, 3)

<u>One-one vertices.</u>

$G_1 \Rightarrow a \rightarrow \deg a = 2$ connected with (3, 3)

$G_2 \Rightarrow V_1 \rightarrow \deg V_4 = 2 \rightarrow (3, 3)$

$\boxed{a \rightarrow V_4}$

$\deg b = 3$ connected $(2, 2, 3)$

$V_2 - (2, 2, 3) \Rightarrow \boxed{b \rightarrow V_2}$

$\deg c = 2$ connected $(3, 3)$

$V_1 \rightarrow (3, 3)$ $\boxed{c \rightarrow V_1}$

$\deg d = 3$ connected $(2, 2, 3)$ $\boxed{d \rightarrow V_3}$

$V_3 = (2, 2, 3)$.

One - one edges

	G_1		G_2
(i) a to b	~~0~~ 1	v_4 to v_2	1
a to c	0	v_4 to v_1	0
a to d	1	v_4 to v_3	1
(ii) b to c	1	v_2 to v_1	1
b to d	1	v_2 to v_3	1
(iii) c to d	1	v_1 to v_3	1

(i) same

(ii) same

(iii) same

Adjacency matrix

$$G_1 = \begin{array}{c|cccc} & a & b & c & d \\ \hline a & 0 & 1 & 0 & 1 \\ b & 1 & 0 & 1 & 1 \\ c & 0 & 1 & 0 & 1 \\ d & 1 & 1 & 1 & 0 \end{array}$$

$$G_2 = \begin{array}{c|cccc} & v_4 & v_2 & v_1 & v_3 \\ \hline v_4 & 0 & 1 & 0 & 1 \\ v_2 & 1 & 0 & 1 & 1 \\ v_1 & 0 & 1 & 0 & 1 \\ v_3 & 1 & 1 & 1 & 0 \end{array}$$

∴ Adjacent matrices are same.

Therefore, two graphs are isomorphic.

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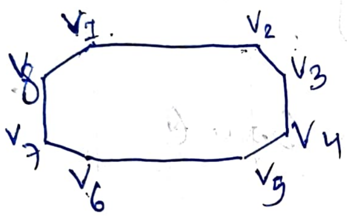
~~Adjacent vertices~~

Bi-Partite

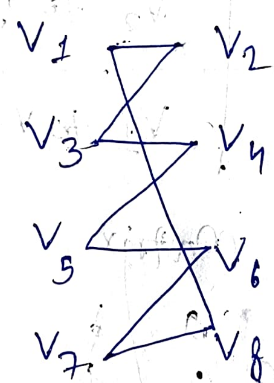
Consider $V_1 = \{v_1, v_3, v_5, \ldots\}$

$V_2 = \{v_2, v_4, v_6 \ldots\}$

such that $V_1 \cap V_2 = \phi$ that is for any edge the end points of e_j should not be in the same set.



→



$v_1 < \begin{matrix} v_2 \\ v_8 \end{matrix}$

$v_3 < \begin{matrix} v_2 \\ v_4 \end{matrix}$

$v_5 < \begin{matrix} v_4 \\ v_6 \end{matrix}$

$v_7 < \begin{matrix} v_8 \\ v_6 \end{matrix}$

→ Each vertex in the first set V_1 is connected each vertex in the second set V_2.

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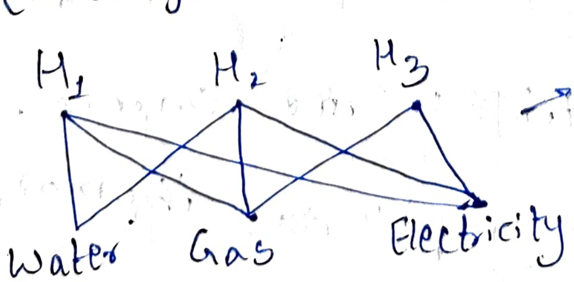
② Draw a graph representing the problem of 3 houses & 3 utilities water, Gas and electricity

A) $V_1 = \{H_1, H_2, H_3\}$

$V_2 = \{$water, gas, electricity$\}$



← Completed Bi-partite graph.

③ Is there a graph with vertices {1, 3, 3, 3, 4, 4, 5, 6}

A) odd no of vertices must be even.
odd vertices are 1, 3, 3, 3, 5. the number is not even, therefore there is no graph.

④ Identify the no of edges in a graph with 10 vertices each of degree '6'.

A) $\sum_{i=1}^{10} deg(v_i) = 2|E|$

$\sum_{i=1}^{10} deg(v_i) = 2|E|$

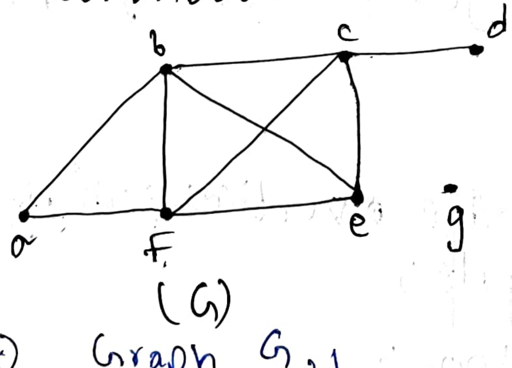
$deg(v_1) + deg(v_2) + \dots + deg(v_{10}) = 2|E|$

$6 + 6 + \dots + 6 = 2|E|$

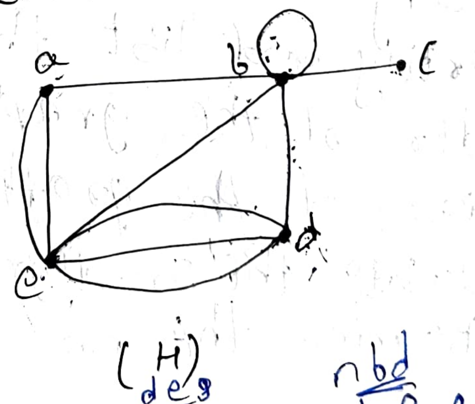
$\frac{60}{2} = |E|$

$|E| = 30$ edges (by hand shaking theorem).

⑤ What are the degrees and what are the neighbourhoods of the vertices in the graph G & H.



(G)



(H)

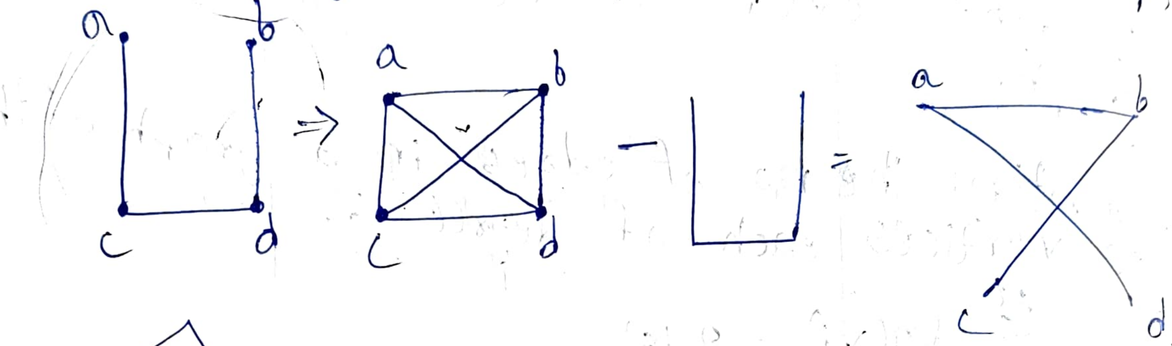
A) Graph G

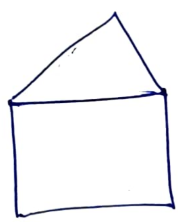
	deg	nbd
a	2	b, f
b	4	a, c, e, f
c	4	b, d, e, f
d	1	c
e	3	b, c, f
f	3	b, c, e
g	0	0

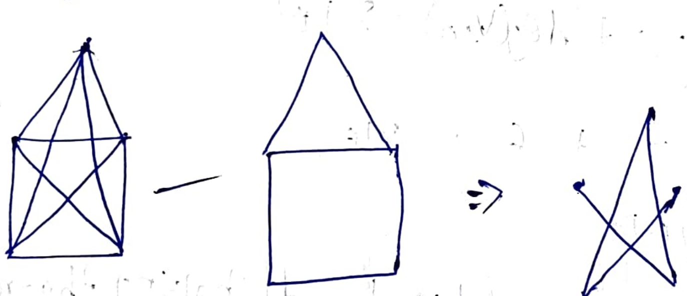
	deg	nbd
a	3	b, e, e
b	4+2	a, e, d, c, loop
c	1	b
d	4	b, e, e, e
e	6	a, a, b, d, d, d

⑥ Draw the complement graph of a given simple graph.

A) Compliment graph = Complete graph - given graph







⑦ Consider an undirected graph.

(i) Identify and list the vertices of the graph

(ii) Identify and list the edges of the graph

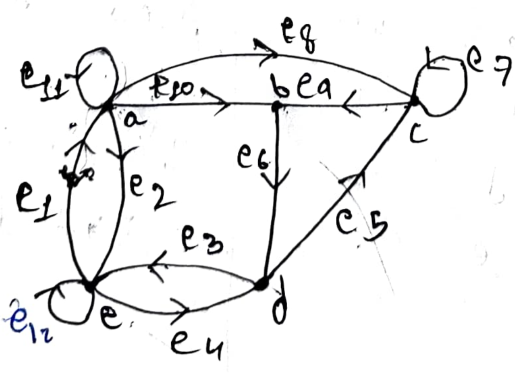
(iii) Order of the graph

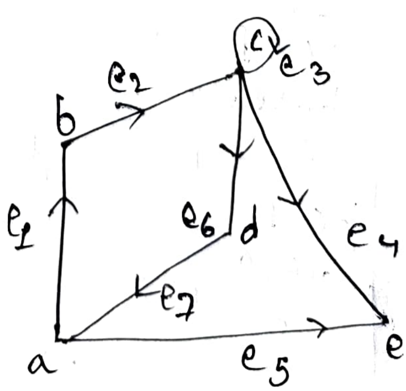
(iv) Size of the graph.

(v) Calculate the In-degree of the vertices & out-degree

(vi) Determine the no of cycles

(vii) " " no of loops





A) vertices = a, b, c, d, e

edges = $e_1, e_2, e_3, \ldots e_{12}$

Order = $|V| = 5$

Size = $|E| = 12$

loops = 3

cycles = ede, abca, bcdb, abdea, aea

	In-degree (deg^-)	Out-degree (deg^+)
a	1 + 1	3 + 1
b	2	1
c	2 + 1	1 + 1
d	2	2
e	2 + 1	2 + 1

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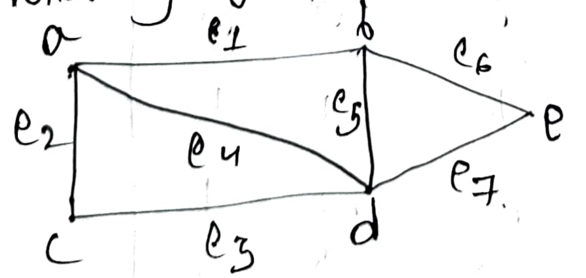
1) In directed graph the In-degree of a vertex is 3 and out-degree is 2 how many edges are incident to vertex A. By hand-shaking theorem for directed graph $\sum deg^-(v_i) + deg^+(v_i) = |E|$.

A) $3 + 2 = |E|$

$|E| = 5$.

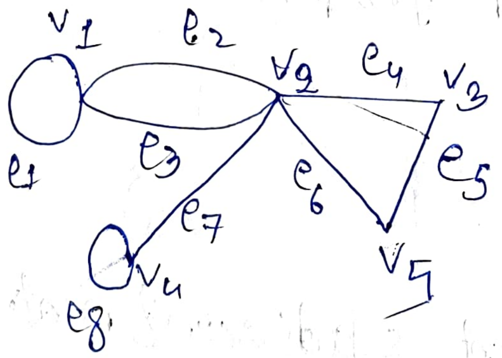
2) Obtain the incident & adjacent matrix of the following graph.



Adjacent Matrix

$$A_G = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \end{array} \begin{array}{c} \begin{array}{ccccc} a & b & c & d & e \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

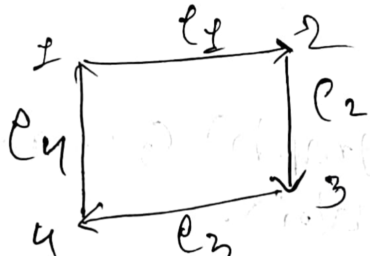
$$I_G = \begin{array}{c} \\ a \\ b \\ c \\ d \\ e \end{array} \begin{array}{c} \begin{array}{ccccccc} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \end{array}$$



$$A_G = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array} \begin{array}{c} \begin{array}{ccccc} v_1 & v_2 & v_3 & v_4 & v_5 \end{array} \\ \begin{bmatrix} 2 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix} \end{array}$$

$$I_G = \begin{array}{c|cccccccc} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \\ \hline v_1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ v_2 & 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ v_3 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ v_4 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ v_5 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{array}$$

3



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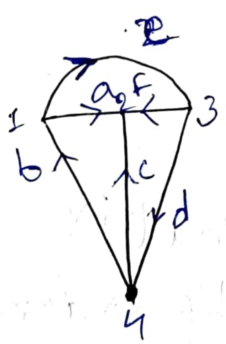
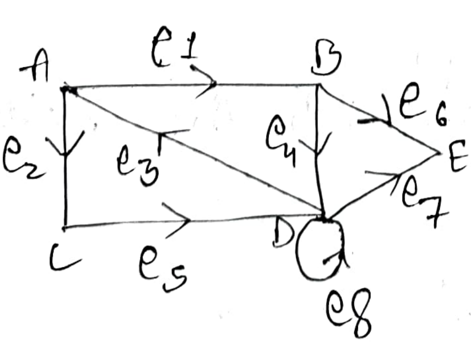


$$A_G = \begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 0 & 1 & 1 & 1 \\ 2 & 1 & 0 & 1 & 1 \\ 3 & 1 & 1 & 0 & 1 \\ 4 & 1 & 1 & 1 & 0 \end{array}$$

$$I_G = \begin{array}{c|cccccc} & a & b & c & d & e & f \\ \hline 1 & 1 & -1 & 0 & 0 & 1 & 0 \\ 2 & -1 & 0 & -1 & 0 & 0 & -1 \\ 3 & 0 & 0 & 0 & 1 & -1 & 1 \\ 4 & 0 & 1 & 1 & -1 & 0 & 0 \end{array}$$



$$A_G = \begin{array}{c|ccccc} & A & B & C & D & E \\ \hline A & 0 & 1 & 1 & 1 & 0 \\ B & 1 & 0 & 0 & 1 & 1 \\ C & 1 & 0 & 0 & 1 & 0 \\ D & 1 & 1 & 1 & 2 & 1 \\ E & 0 & 1 & 0 & 1 & 0 \end{array}$$

$$I_G = \begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array} \begin{array}{cccccccc} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \\ \end{array}$$

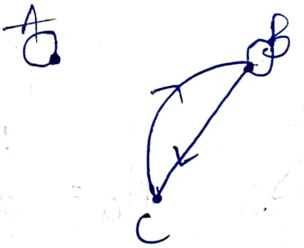
$$I_G = \begin{array}{c|cccccccc} & e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \\ \hline A & 1 & 1 & -1 & 0 & 0 & 0 & 0 & 0 \\ B & -1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ C & 0 & -1 & 0 & 0 & 1 & 0 & 0 & 0 \\ D & 0 & 0 & 1 & -1 & -1 & 0 & 1 & 1 \\ E & 0 & 0 & 0 & 0 & 0 & -1 & -1 & 0 \end{array}$$

Session - 24

1) Draw the diagram (directed graph) G corresponding to the Adjacency matrix.

A)

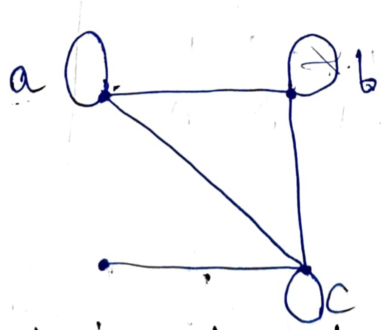
$$A_G = \begin{array}{c} \\ A \\ B \\ C \end{array} \begin{array}{c} \begin{array}{ccc} A & B & C \end{array} \\ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \end{array}$$



2.) Draw a graph with adjacency matrix with respective the ordering of the vertices a, b, c, d.

$$A_G = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

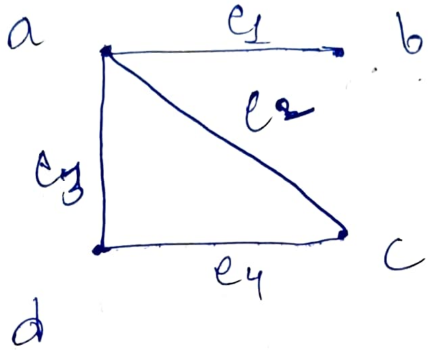
A)



3.) Draw the undirected graph with the incidence matrix

$$I_G = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{cccc} e_1 & e_2 & e_3 & e_4 \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \end{array}$$

A)

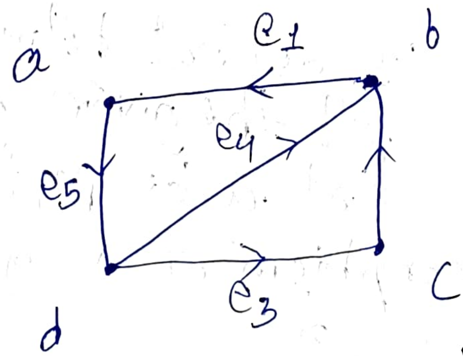


* For Incidence matrix we have to consider column wise.

4) Draw the diagram with Incidence matrix

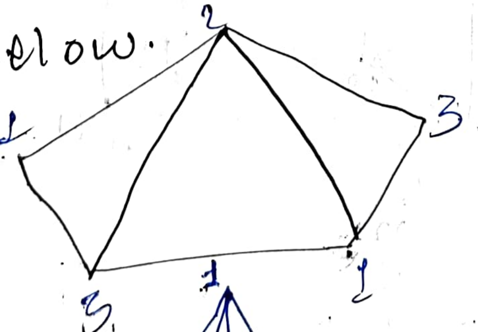
$$I_G = \begin{array}{c} \\ a \\ b \\ c \\ d \end{array} \begin{array}{c} \begin{array}{ccccc} e_1 & e_2 & e_3 & e_4 & e_5 \end{array} \\ \begin{bmatrix} -1 & 0 & 0 & 0 & 1 \\ 1 & -1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 & -1 \end{bmatrix} \end{array}$$

A)

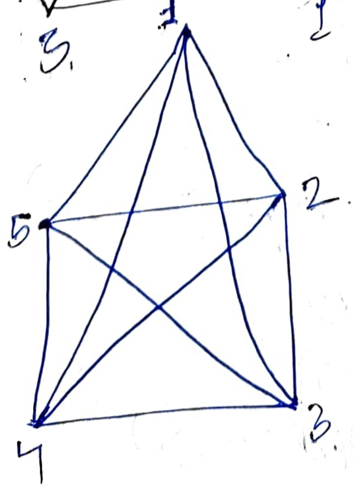


5) Find the chromatic number for the graph given below.

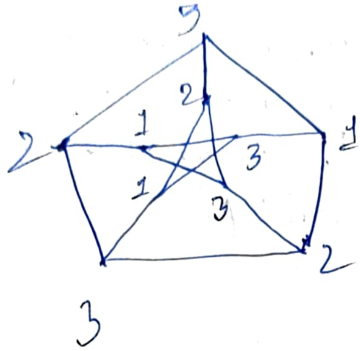
A)



$\chi(G) = 3$



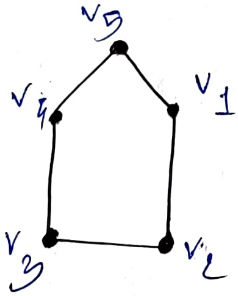
$\chi(G) = 5$

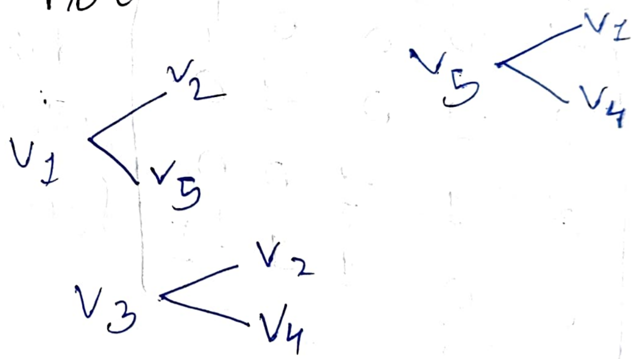


$\chi(G) = 3$

6) Test whether the given graph is complete Bi-partite or not.

A)



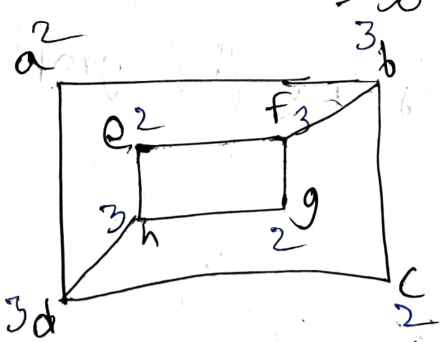


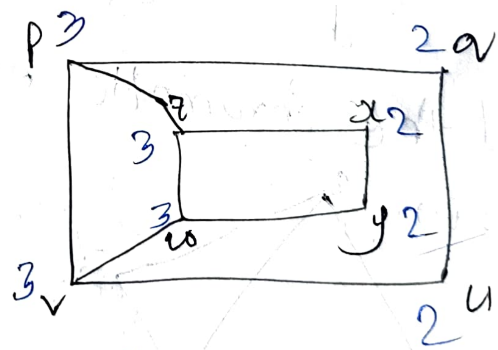
$V_1 = \{v_1, v_3\}$

$V_2 = \{v_2, v_4\}$

v_5 does not belongs to any set the given graph is not a Bi-partite.

7) Examine the Isomorphism of a graph.





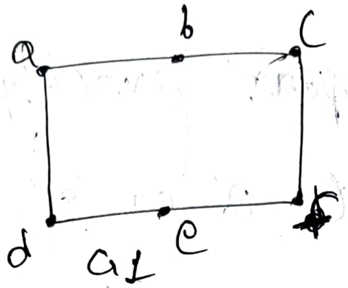
	G_1	G_2
no of vertices	8	8
no of edges	10	10
degree sequence	(2,2,2,2,3,3,3,3)	(2,2,2,2,3,3,3,3)

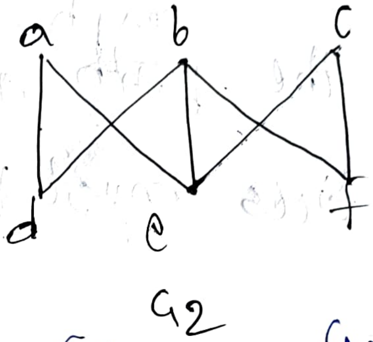
One-one correspondance for vertices

G_1 - deg(a) = 2 connected (3,3)

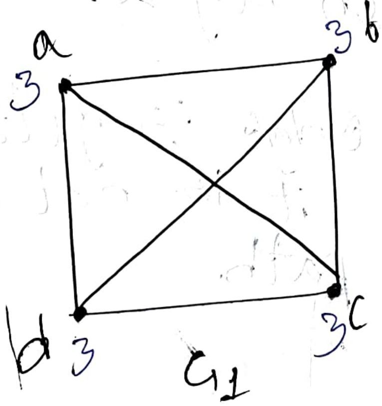
G_2 - deg(u) = 2 not connected (3,3)

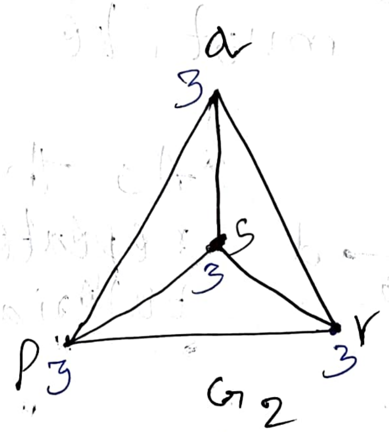
∴ The given graphs are not isomers.





	G_1	G_2
No of vertices	6	6
No of edges	6	7





	G_1	G_2
No of vertices	4	4
No of edges	6	6
degree sequence	(3,3,3,3)	(3,3,3,3)

one-one correspondance

deg(a) = 3 connected to (3,3)

deg(a) = 3 connected to (3,3)

all are connected to 3,3

$$A_G = \begin{array}{c c} & \begin{array}{cccc} a & b & c & d \end{array} \\ \begin{array}{c} a \\ b \\ c \\ d \end{array} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{array} \qquad A_G = \begin{array}{c c} & \begin{array}{cccc} p & q & r & s \end{array} \\ \begin{array}{c} p \\ q \\ r \\ s \end{array} & \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \end{array}$$

∴ Both the graphs are isomers.

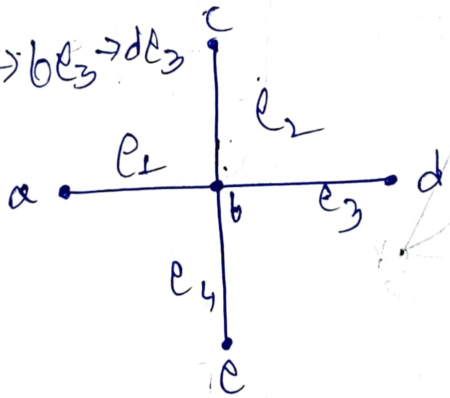
Eulerian path:-

* Every edges of the ~~path~~ graph appears exactly once in the path.
* No of oddvertices must be, '0' or '2'.

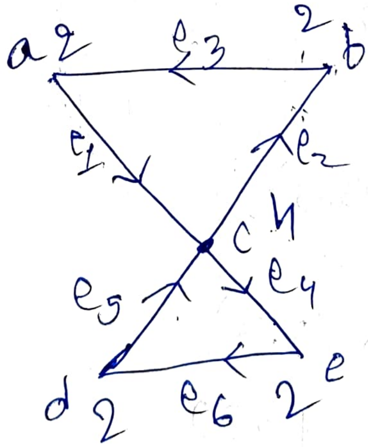
Eulerian circuit:-

* The circuit which contains every edge of the graph exactly once.
* Each vertices must be even degree.

$a e_1 \rightarrow b e_3 \rightarrow d e_3$

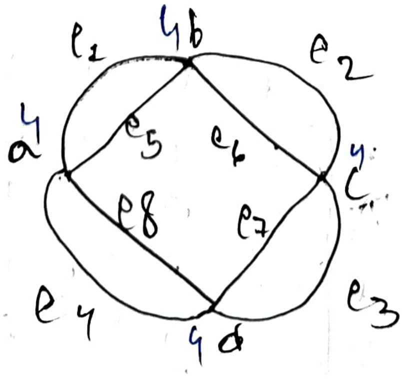


* As the edge is getting repeated it is not a Eulerian path.



* It is a Eulerian path as all the edges follow unique direction. Also Eulerian circuit.

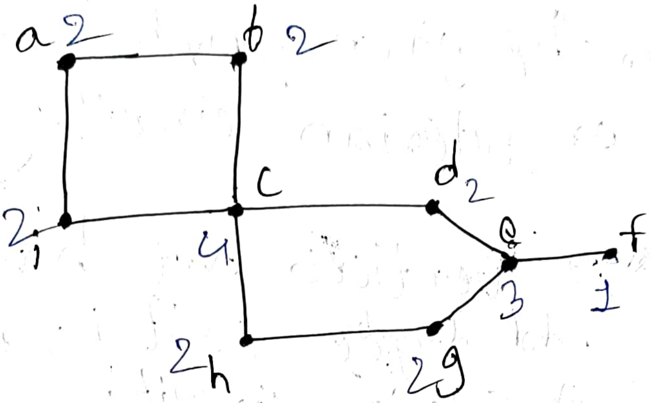
$e\,e_6 - d e_5 - c e_2 - b e_3 - a e_1 - c e_4 - e$



Degree of each vertex is even.

Eulerian Path ⇒ $a e_1 - b e_2 - c e_3 - d e_4 - a e_5 - b e_6 - c e_7 - d e_8$

Eulerian circuit also follows same path.



As 2 vertices are having odd degree it is not a Eulerian circuit.

EP:- f - e - d - c - b - a - i - c - h - g - e.

∴ EP is possible.

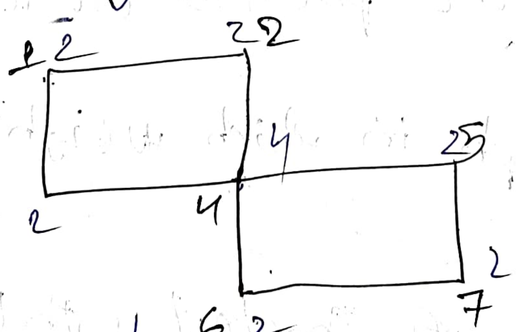
Hamiltonion Path (vertices)

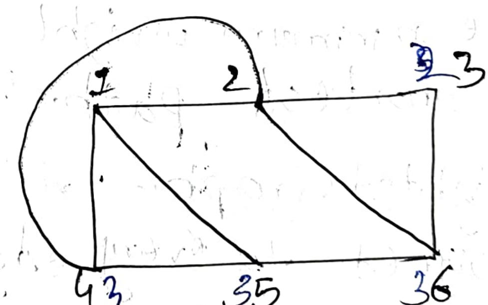
→ Every vertex of the graph appears exactly once in the path.

• Hamiltonion Circuit

The circuit which contains every vertex of the graph exactly once.

Ex:-





Not a hamiltonion circuit as the initial & terminal vertices are not co-inciding.

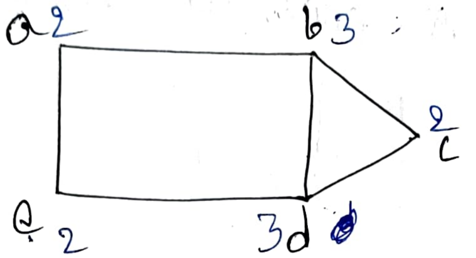
→ Hamiltonion Path exist

3124576 to 4

H.P:- 4 1 2 5 6 3

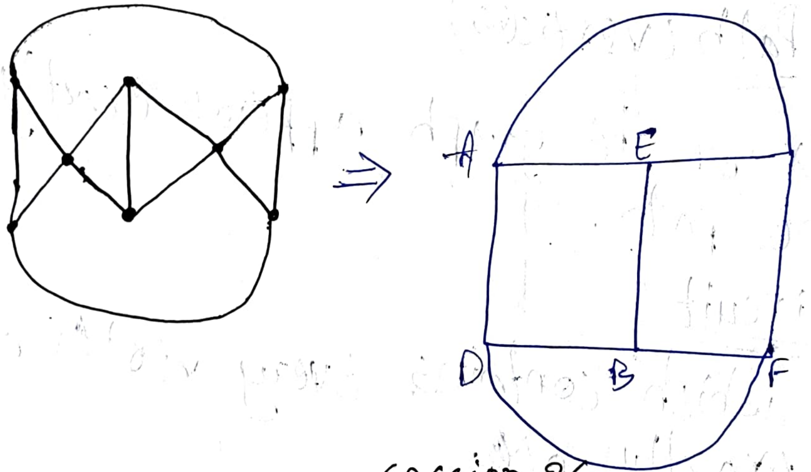
H.C:- 1 2 3 6 5 4 1

Determine whether the given graph has hamiltonion circuit or Eulerian circuit.



* 2 vertices are having odd degree so it is not a Eulerian circuit.

* H.C = c b a e d c



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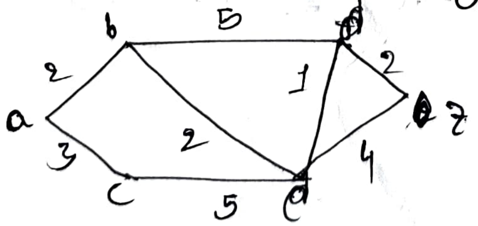
The minimum weight of the path is the length of shortest path.

<u>Weighted Graph</u>:- A graph in which weights are assigned to every edge.

Q. Determine the length of shortest path between A & z in the weighted graph.



a - b - d - z = 2+5+2 = 9

a - b - e - d - z = 2+2+1+2 = 7

a - b - e - z = 2+2+4 = 8

a - c - e - z = 3+5+4 = 12

a - c - e - d - z = 3+5+1+2 = 11

a - c - e - b - d - z = 3+5+2+5+2 = 17.

∴ min weight is 7. The shortest path is a - b - e - d - z.

Dijkstra's Algroriithm:-

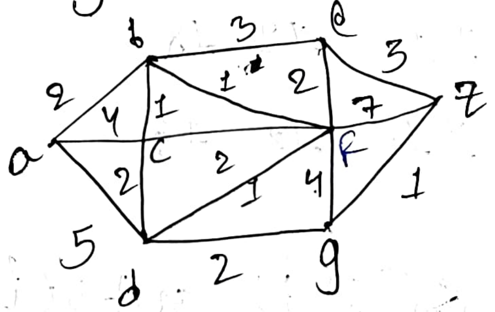
Step 1:- Write all the vertices from A to Z.

Step 2:- Take initial value 'a' as zero..

Step 3:- If any edge is not connected consider as ~~zero~~. infinity.

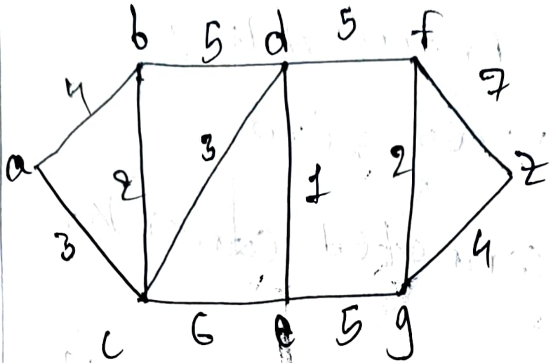
* Suppose an edge from B to C whose weight is 'k' and the initial value of B is 'n'. Therefore the value of C is n+k. (Take the min value compared to the previous one).

Apply Dijkstra's Algorithm to find the length of a shortest path between A & Z in the given weighted graph.



* Shortest path is
a-b-f-d-g-z = 2+1+1+2+1
= 7

a	b	c	d	e	f	g	z
0	2	4	5	∞	∞	∞	∞
0	2	3	5	5	3	∞	∞
0	2	3	5	5	3	∞	∞
0	2	3	4	5	3	7	10
0	2	3	4	5	3	6	10
0	2	3	4	5	3	6	8
0	2	3	4	5	3	6	7



a	b	c	d	e	f	g	z
0	4	3	∞	∞	∞	∞	∞
0	4	3	6	9	∞	∞	∞
0	4	3	6	9	∞	∞	∞
0	4	3	6	7	11	∞	∞
0	4	3	6	7	11	12	∞
0	4	3	6	7	11	12	18
0	4	3	6	7	11	12	16

Spannig Tree

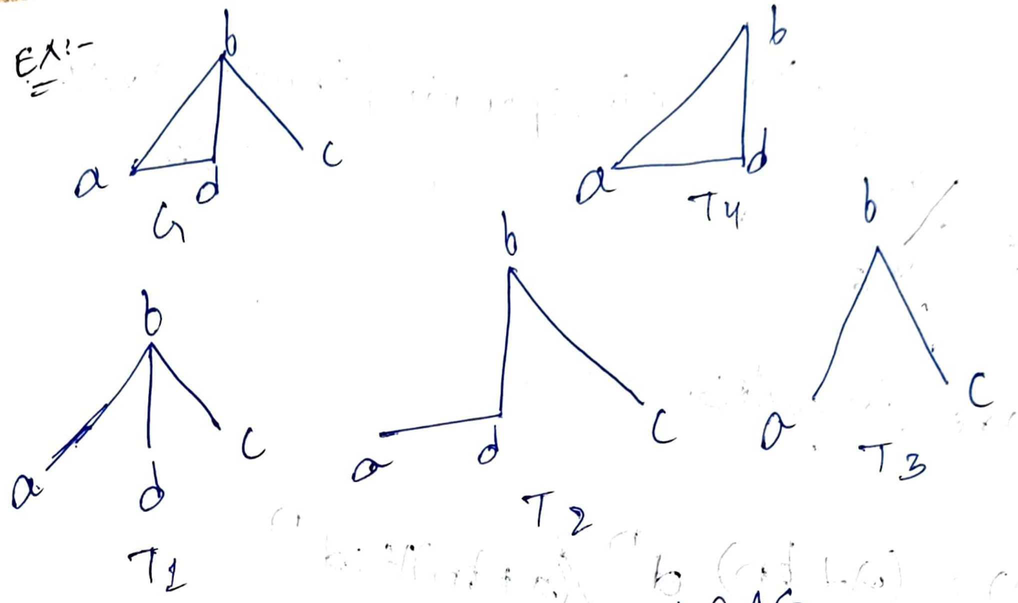
→ It is developed from graph 'G' is subset of 'G' which cover all the vertices using mi number.

$v(G) = v(T)$.

Note:-

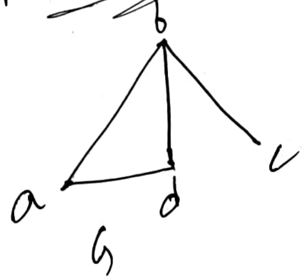
* It covers all the vertices.
* cannot have any cycle.
* Have (n-1) edges.

Ex:-



* T_1 & T_2 are spanning trees
* T_3 & T_4 are not

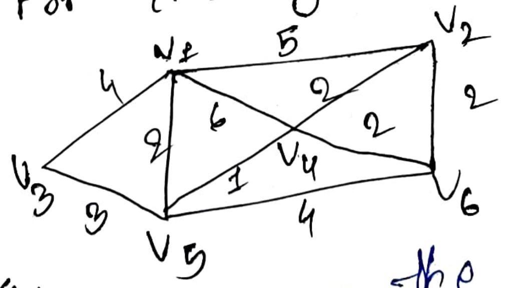
Minimum Spanning Tree



Let 'G' be a weighted undirected graph and t_1 be a spanning tree is called min spanning tree if it is than other spanning trees of graph 'G'.

Session - 27

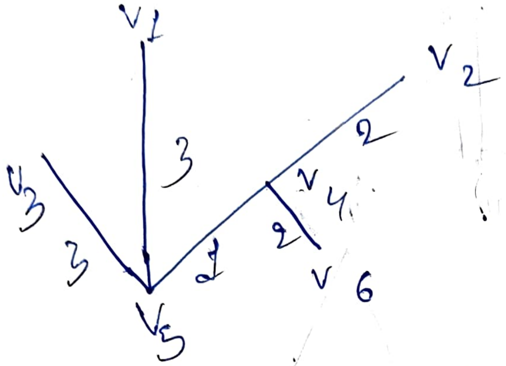
* Construct min spanning tree for the given graph using Kruskal's



Step 1:- Arrange the edges in ascending order.

V_4V_5 (1) V_4V_2 (2) V_4V_6 (2) V_2V_6 (2) V_1V_5 (2) V_3V_5 (3) V_1V_2 (4)

V_5V_6 (4) V_1V_2 (5) V_1V_4 (6)



Min spanning tree $= 3+2+1+2+$
$= 10$

→ There are 10 cycles.

If $f(n) = (a+bn)\,d^n \,/\, (a+bn)n^k \cdot d^n$

d - not root $\Rightarrow a_n = (P_0 + P_1 n)\,d^n$

d - root $\quad a_n = (P_0 + P_1 n)\,n\,d^n$

d - root/twice $\quad a_n = (P_0 + P_1 n)\,n^2\,d^n$.